

Multiple Regression Analysis

多元回归分析

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + u$$

4. Further Issues

进一步的问题

Chapter Outline 本章大纲

- Effects of Data Scaling on OLS Statistics

数据的测度单位换算对OLS统计量的影响

- More on Functional Form

对函数形式的进一步讨论

Lecture Notes 课堂笔记

■ Effects of Redefining variables

重新定义变量的影响

- Estimated coefficients 估计系数
- R squared R 平方
- t statistics t 统计量

■ Functional form 函数形式

- Logarithmic form 对数函数形式
- Models with Quadratics 含二次式的模型
- Models with interaction terms 含交叉项的模型

Redefining Variables

重新定义变量

- Why would we want to do so?

为什么我们想这样做？

- Often, data scaling is used to reduce the number of zeros after a decimal point in an estimated coefficient, so that the results appear prettier.

数据测度单位变换经常被用于减少被估参数小数点后的零的个数，这样结果更好看一些。

- Since this is mainly an action of decoration, we expect nothing essential should change.

既然这样做主要为了好看，我们希望本质的东西不改变。

Redefining Variables: An example

重新定义变量：一个例子

- Consider a model relating infant birth weight to cigarette smoking and family income:

以下模型反映了婴儿出生体重与孕妇吸烟量和家庭收入之间的关系：

$$(1) \text{ } bwght = \hat{\beta}_0 + \hat{\beta}_1 \text{cigs} + \hat{\beta}_2 \text{faminc}$$

- Consider the following rescaling:

考虑如下单位变换：

(2) Birth weight is changed from ounces to pounds

出生体重单位由盎司变为磅

(3) Number of cigarettes is changed to packs of cigarettes

香烟的支数变为包数

- The estimation results are presented in the following table.

估计结果列于下表

Table 6.1

Y (column)	(1)bwght(盎司)	(2)bwghtlbs(磅)	(3) bwght(盎司)
X (rows)			
cigs(条)	-0.4634 (0.0916)	-0.0289 (0.0057)	--
packs(包)	--	--	-9.268 (1.832)
faminc	0.0927 (0.0292)	0.0058 (0.0018)	0.0927 (0.0292)
Intercept	116.794 (1.049)	7.3109 (0.0656)	116.974 (1.049)
Observations	1388	1388	1388
R-squared	0.0298	0.0298	0.0298
SSR	557,485.51	2177.5778	557.485.51
SE	20.063	1.2539	20.063

Impact of changing the scale of the dependent variable 改变被解释变量测度单位的影响

- Since 1lbs = 16 oz, the dependent variable is transformed by dividing 16.

因为1磅=16盎司，被解释变量被除以16。

$$bwght / 16 = \hat{\beta}_0 / 16 + (\hat{\beta}_1 / 16)cigs + (\hat{\beta}_2 / 16)faminc$$

- We compare columns (1) and (2).

比较第1列与第2列。

- The estimated coefficients in (1)/16 = those in (2).

(1)中被估参数/16 = (2)中被估参数

- The standard errors of estimated coefficients in (1)/16 = those in (2)

(1)中被估参数的标准差/16 = (2)中被估参数的标准差

Impact of changing the scale of the dependent variable 改变被解释变量测度单位的影响

- The t statistics in (1) and (2) are identical.

(1)和(2)中 t 统计量相同

- The R squared values are identical.

R平方相同

- $SSR \text{ in (1) } / (16 * 16) = SSR \text{ in (2)}$

(1)中SSR/ (16*16) = (2)中SSR

- $SE \text{ (standard error) in (1) } / 16 = SE \text{ in (2)}$ $\hat{\sigma}$

(1)中SE(标准差)/16 = (2)中SE

Impact of changing the scale of the independent variable 改变解释变量测度单位的影响

- Now number of cigarettes is changed to packs of cigarettes. 现在香烟数量单位变为包。

$$bwght = \hat{\beta}_0 + (\hat{\beta}_1 * 20)(cigs / 20) + \hat{\beta}_2 faminc$$

- Now compare columns (1) and (3).

现在比较 第(1)列和第(3)列。

- Coefficients estimates and standard errors on faminc and intercept are the same.

变量faminc系数和截距项的估计值和其标准差分析同上。

- Coefficients estimates and standard errors on packs are 20 times larger.

packs的系数估计值和标准差变为20倍。

Impact of changing the scale of the independent variable 改变解释变量测度单位的影响

- The t statistics are identical.

t 统计量相同

- The R squared values are identical.

R 平方相同

- The SSR are identical.

SSR相同

- The SE are identical.

SE相同

Redefining Variables

重新定义变量

- Changing the scale of the y variable will lead to a corresponding change in the scale of the coefficients and standard errors, so no change in the significance or interpretation

改变变量 y 的测度单位会导致系数和标准差相应的改变，所以解释变量系数显著性和对其解释没有改变。

- Changing the scale of one x variable will lead to a change in the scale of that coefficient and standard error, so no change in the significance or interpretation on this variable and other variables.

改变一个变量 x 的测度单位会导致该变量系数和标准差的相应改变，所以所有解释变量显著性和对其解释没有改变。

思考题

- 原模型关于消费（**cons**，千美元）和家庭子女数量（**kids**）的回归方程如下：

$$\ln(\text{cons}) = \beta_0 + \beta_1 \text{kids} + \beta_2 x + \varepsilon,$$

- 如果**cons**改成以千人民币计算，请问以下变量会有什么变化？
 - (1) $\hat{\beta}_1$ 、参数标准差、t值
 - (2) R^2 、调整后 R^2
 - (3) SSR、 $\hat{\sigma}$

Redefining Variables

重新定义变量

- If the dependent variable appears in logarithm form, changing the unit of measurement of the dependent variable does not affect any of the slope coefficient.

如果被解释变量以对数形式出现，改变被解释变量度量单位对任何斜率系数没有影响。

- This follows from $\log(cy) = \log(c) + \log(y)$, rescaling y will result in changes to the intercept but not the slope coefficients.

来自 $\log(cy) = \log(c) + \log(y)$ ，改变 y 测度单位将改变截距，不改变斜率系数。

Beta Coefficients Beta系数

- Consider a sample regression function of the following form:

考虑如下形式的样本回归方程：

$$\hat{y} = 200 + 20,000x_1 + 0.2x_2$$

- Can we say that x_1 is the most important variable?

我们能说 x_1 是最重要的变量吗？

- Now look at the units of each variable:

现在，查看以下各个变量的单位：

- y in dollars y 单位： 美元
- x_1 in cents x_1 单位： 美分
- x_2 in thousands x_2 单位： 千美元

Beta Coefficients Beta系数

- What problem does the above example reveal?

上例揭示了什么问题？

- The magnitudes of the estimated coefficients are not comparable.

被估计系数的大小是不可比较的。

- A related problem is when the magnitudes of variables differ too much, the round-off error can be serious in regression calculations.

一个相关的问题是，当变量大小差别过大时，在回归中因运算近似而导致的误差会比较大。

Beta Coefficients Beta系数

- Occasionally you'll see reference to a “standardized coefficient” or “beta coefficient” which has a specific meaning

有时，我们会看见“标准化系数”或“Beta系数”，这些名称有着特殊的意义

- Idea is to replace y and each x variable with a standardized version – i.e. subtract mean and divide by standard deviation

使用Beta系数是因为有时我们把 y 和各个 x 替换为标准化版本——也就是，减去均值后除以标准离差。

- Coefficient reflects standard deviation of y for one standard deviation change in x

系数反映对于一单位 y 标准离差对一单位 x 标准离差的影响。 16

Beta Coefficients Beta系数

The normal form of the sample regression function is

样本回归方程的标准形式是

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \dots + \hat{\beta}_k x_{ik},$$

Standard y_i and x_{ij} using $z_y = \frac{y_i - \bar{y}}{\hat{\sigma}_y}$, $z_{x_j} = \frac{x_{ij} - \bar{x}_j}{\hat{\sigma}_j}$, $j = 1, \dots, k$.

标准化 y_i 和 x_{ij} , $z_y = \frac{y_i - \bar{y}}{\hat{\sigma}_y}$, $z_{x_j} = \frac{x_{ij} - \bar{x}_j}{\hat{\sigma}_j}$, $j = 1, \dots, k$.

Now regress z_y on z_{x_j} to get 现在将 z_y 向 z_{x_j} 回归得到

$$\hat{z}_y = \hat{b}_1 z_{x_1} + \hat{b}_2 z_{x_2} + \dots + \hat{b}_k z_{x_k}$$

Notice (2) do not have an intercept term

注意(2)没有截距项

Now, what is the relation between $\hat{b}_j$ and $\hat{\beta}_j$?

现在, $\hat{b}_j$ 与 $\hat{\beta}_j$ 的关系如何?

Beta Coefficients Beta系数

Since

$$\frac{\hat{y}_i - \bar{y}}{\hat{\sigma}_y} = \frac{\hat{\sigma}_1}{\hat{\sigma}_y} \hat{\beta}_1 \frac{x_{i1} - \bar{x}_1}{\hat{\sigma}_1} + \frac{\hat{\sigma}_2}{\hat{\sigma}_y} \hat{\beta}_2 \frac{x_{i2} - \bar{x}_2}{\hat{\sigma}_2} + \dots + \frac{\hat{\sigma}_k}{\hat{\sigma}_y} \hat{\beta}_k \frac{x_{ik} - \bar{x}_k}{\hat{\sigma}_k}$$

we have

$$\hat{b}_j = \frac{\hat{\sigma}_j}{\hat{\sigma}_y} \hat{\beta}_j, j = 1, 2, \dots, k$$

are traditionally called standardized coefficients or beta coefficients. Meaning, if x_{ij} changes one standard deviation, then y_i changes $\hat{b}_j$ standard deviation.

传统上 $\hat{b}_j$ 被称为标准化系数或 β 系数。

意思是，如果 x_{ij} 改变一单位标准离差，则 y_i 改变 $\hat{b}_j$ 单位标准离差。

Functional Form 函数形式

- OLS can be used for relationships that are not strictly linear in x and y by using nonlinear functions of x and y – will still be linear in the parameters

OLS也可以用在 x 和 y 不是严格线性的情况，通过使用非线性方程，使得关于参数仍为线性。

- Can take the natural log of x , y or both

可以取 x , y （一个或全部）的自然对数

- Can use quadratic forms of x

可以用 x 的平方形式

- Can use interactions of x variables

可以用 x 的交叉项

Interpretation of Log Models

对数模型的解释

■ If the model is $\ln(y) = \beta_0 + \beta_1 \ln(x) + u$

如果模型是 $\ln(y) = \beta_0 + \beta_1 \ln(x) + u$

■ β_1 is the elasticity of y with respect to x

β_1 是 y 对于 x 的弹性

■ If the model is $\ln(y) = \beta_0 + \beta_1 x + u$

如果模型是 $\ln(y) = \beta_0 + \beta_1 x + u$

■ β_1 is approximately the percentage change in y given a 1 unit change in x , often called semi-elasticity.

β_1 近似是，给定一单位 x 的改变， y 的百分比变化，常被称为半弹性。

Why use log models?

为什么使用对数模型？

- The slope coefficients on Logged variables are invariant to the scale of the variables.

取对数后变量的斜率系数，不随变量测度单位改变。

- They give a direct estimate of elasticity if both regressor and regressand have taken logs.

如果回归元和回归子都取对数形式，斜率系数给出对弹性的一个直接估计。

- For models with $y > 0$, the conditional distribution is often heteroskedastic or skewed, while $\ln(y)$ is much less so. The distribution of $\ln(y)$ is more narrow, limiting the effect of outliers.

对于 $y > 0$ 的模型，条件分布经常偏斜或存在异方差，而 $\ln(y)$ 就小多了，所以 $\ln(y)$ 的分布窄多了，限制了异常（或极端）观测值(outliers)的影响。

Some Rules of Thumb

一些经验法则

- What types of variables are often used in log form? 什么类型的变量经常用对数形式?
 - Dollar amounts that must be positive, wages, salaries, firm sales, and firm market values.
肯定为正的钱数：工资，薪水，企业销售额和企业市值。
 - Very large variables, such as population, total number of employees, school enrollment, etc.
非常大的变量：如人口，雇员总数和学校注册人数等。

Some Rules of Thumb

一些经验法则

- What types of variables are often used in level form?
什么类型的变量经常用水平值形式?
 - Variables measured in years, e.g., education, experience, tenure, age
用年测量的变量：如教育年限，工作经历，任期年限和年龄
- Variables, that can appear either in log or in level: 可以以水平值或对数形式出现的变量：
 - Variables that are a proportion or percent: unemployment rate, the participation rate in a pension, etc.
比例或百分比变量：失业率，养老金参与率等。

Limitations of Logs

对数形式的限制

- It cannot be used if a variable takes on zero or negative values.

一个变量取零或负值，则不能使用对数。

- In cases when y is nonnegative but can take 0, $\log(1+y)$ is sometimes used.

如果 y 非负但可以取零，则有时使用 $\log(1+y)$ 。

- Using $\log(1+y)$ and then interpreting the estimates as if the estimates were $\log(y)$ is acceptable when the data on y are not dominated by zeros.

当数据并非多数为零时，使用 $\log(1+y)$ 估计，并且假定变量为 $\log(y)$ ，解释所得的估计值，是可以接受的。

Cautions in using Logs

慎重使用对数形式

- Notice that when y is in log form, it is more difficult to predict the original variables, since the original model allow us to predict $\log(y)$ instead of y .

注意到，当 y 取对数形式时，更难以预测原变量的值，因为原模型允许我们预测 $\log(y)$ 而不是 y 。

Cautions in using Logs

慎重使用对数形式

Consider $\log \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1$.

考虑 $\log \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1$.

$x_1 \uparrow 1 \text{ unit}$, $y \uparrow$ by $(100\hat{\beta}_1)\%$,

当 x_1 变化1个单位时, y 变化 $(100\hat{\beta}_1)\%$,

if the initial level of y is y_0 , the expected value of y is $y_0 (1 + \hat{\beta}_1)$.

如果 y 当前值为 y_0 , 那么预计 y 的值将是 $y_0 (1 + \hat{\beta}_1)$ 。

Quadratic Models

含二次式的模型

- For a model of the form $y = \beta_0 + \beta_1 x + \beta_2 x^2 + u$ we can't interpret β_1 alone as measuring the change in y with respect to x , we need to take into account β_2 as well, since

对于形式为 $y = \beta_0 + \beta_1 x + \beta_2 x^2 + u$ 的模型，我们不能单独将 β_1 解释为关于 x ， y 变化的度量，我们需要将 β_2 也考虑进来，因为

$$(1) \quad \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2$$

$$(2) \quad \Delta \hat{y} \approx (\hat{\beta}_1 + 2\hat{\beta}_2 x) \Delta x, \text{ so}$$

$$(3) \quad \frac{\Delta \hat{y}}{\Delta x} \approx \hat{\beta}_1 + 2\hat{\beta}_2 x$$

Quadratic Models

含二次式的模型

- If one is interested in calculating the predicted changes in y given a starting value of x and a change in x , one can directly use (1).

如果感兴趣的是，给定 x 的初始值和变动，预测 y 的变化，那么可以直接使用（1）。

- In general, we may use the average value of x , or the median, or the lower and upper quantiles to predict y , depending on the question of our interest.

一般来说，我们可以使用 x 的平均值（最常用），中值，或上下四分位数来预测 y ，取决于我们感兴趣的问题。

Quadratic Models

含二次式的模型

In many cases $\hat{\beta}_1 > 0$, but $\hat{\beta}_2 < 0$. For example

$$\widehat{wage} = 3.73 + 0.298 \text{exper} - 0.0061 \text{exper}^2$$

(0.35) (0.041) (0.0009)

在许多情况下 $\hat{\beta}_1 > 0$, 但 $\hat{\beta}_2 < 0$ 。

例如 $\widehat{wage} = 3.73 + 0.298 \text{exper} - 0.0061 \text{exper}^2$

(0.35) (0.041) (0.0009)

The above equation implies that *exper* has a diminishing effect on wage:

$$\frac{\Delta wage}{\Delta exper} \approx 0.298 - 2 * 0.0061 * \text{exper}.$$

以上式子意味着 *exper* 对工资有递减的影响：

$$\frac{\Delta wage}{\Delta exper} \approx 0.298 - 2 * 0.0061 * \text{exper}..$$

Quadratic Models

含二次式的模型

If $x = 0$, wage will increase 0.298 dollar if experience is increased from 0 to 1 year.

If $x = 1$, then the experience from year 1 to year 2 worths an additional 0.286 dollar, and so on.

In this case there will be a value of x that after this point, x will have a negative effect on y .

如果 $x = 0$, 那么experience从0年增加到1年, 工资增加0.298千美元。

如果 $x = 1$, 那么experience从0年增加到1年, 值额外的0.286千美元, 等等。
在这个例子中, 存在一个值, 过了这个值后, x 对 y 有负的影响。

Wage
(1000\$)

7.37

3.73

24.4

exper

Intermediate Econometrics,
SILC



More on Quadratic Models

对含二次式模型的进一步讨论

- Suppose that the coefficient on x is positive and the coefficient on x^2 is negative

假如 x 的系数为正， x^2 的系数为负。

- Then y is increasing in x at first, but will eventually turn around and be decreasing in x

那么， y 首先随 x 上升而上升，但最终转向随 x 上升而下降。

For $\hat{\beta}_1 > 0$ and $\hat{\beta}_2 < 0$ the turning point

will be at $x^* = \left| \hat{\beta}_1 / (2\hat{\beta}_2) \right| = \hat{\beta}_1 / (-2\hat{\beta}_2)$.

对于 $\hat{\beta}_1 > 0$ ， $\hat{\beta}_2 < 0$ ，转向点 $x^* = \left| \hat{\beta}_1 / (2\hat{\beta}_2) \right|$ 。

More on Quadratic Models

对含二次式模型的进一步讨论

- Suppose that the coefficient on x is negative and the coefficient on x^2 is positive

假如 x 的系数为负， x^2 的系数为正。

- Then y is decreasing in x at first, but will eventually turn around and be increasing in x

那么， y 首先随 x 上升而下降，但最终转向随 x 上升而上升。

For $\hat{\beta}_1 < 0$ and $\hat{\beta}_2 > 0$ the turning point

will be at $x^* = \left| \hat{\beta}_1 / (2\hat{\beta}_2) \right| = -\hat{\beta}_1 / (2\hat{\beta}_2)$, which is

the same as when $\hat{\beta}_1 > 0$ and $\hat{\beta}_2 < 0$

对于 $\hat{\beta}_1 < 0$, $\hat{\beta}_2 > 0$,

转向点 $x^* = \left| \hat{\beta}_1 / (2\hat{\beta}_2) \right| = -\hat{\beta}_1 / (2\hat{\beta}_2)$, 与 $\hat{\beta}_1 > 0$, $\hat{\beta}_2 < 0$ 时相同。

Interaction Terms

交叉项

- For a model of the form $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + u$ we can't interpret β_1 alone as measuring the change in y with respect to x_1 , we need to take into account β_3 as well, since
- 对于形式为 $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + u$ 的模型，我们不能单独将 β_1 解释为关于 x_1 ， y 变化的度量，我们需要将 β_3 也考虑进来，因为

Interaction Terms

交叉项

$$\frac{\Delta y}{\Delta x_1} = \beta_1 + \beta_3 x_2,$$

for example, y is the $\log(\text{price})$, x_1 is the number of bedrooms, and x_2 = square foot of the rooms

so to summarize the effect of x_1 on y , we typically evaluate the above at $\bar{x}_2$

例如， y 是 $\log(\text{price})$ ， x_1 是卧室数， x_2 是房间面积（平方英尺）。
所以，要总述 x_1 对 y 的影响，
比较典型地做法是在 $\bar{x}_2$ 处估计上式。

Interaction Terms

交叉项

■ Example 6.3, P191.

$$\text{stndfnl} = \beta_0 + \beta_1 \text{atndrte} + \beta_2 \text{priGPA} + \beta_3 \text{ACT} + \beta_4 \text{priGPA}^2 + \beta_5 \text{ACT}^2 + \beta_6 \text{priGPA} \cdot \text{atndrte} + u.$$

$$\begin{aligned} \widehat{\text{stndfnl}} &= 2.05 - .0067 \text{ atndrte} - 1.63 \text{ priGPA} - .128 \text{ ACT} \\ &\quad (1.36) \quad (.0102) \quad \quad \quad (.48) \quad \quad \quad (.098) \\ &+ .296 \text{ priGPA}^2 + .0045 \text{ ACT}^2 + .0056 \text{ priGPA} \cdot \text{atndrte} \\ &\quad (.101) \quad \quad \quad (.0022) \quad \quad \quad (.0043) \\ n &= 680, R^2 = .229, \bar{R}^2 = .222. \end{aligned}$$

Interaction Terms

交叉项

$$\begin{aligned}\widehat{stndfnl} = & 2.05 - .0067 \textit{atndrte} - 1.63 \textit{priGPA} - .128 \textit{ACT} \\ & (1.36) \quad (.0102) \qquad \qquad (.48) \qquad \qquad (.098) \\ & + .296 \textit{priGPA}^2 + .0045 \textit{ACT}^2 + .0056 \textit{priGPA} \cdot \textit{atndrte} \\ & (.101) \qquad \qquad (.0022) \qquad \qquad (.0043) \\ & n = 680, R^2 = .229, \bar{R}^2 = .222.\end{aligned}$$

- ① Each t statistic is insignificant, we cannot reject $H_0: \beta_1 = 0$, and $\beta_6 = 0$, respectively.
- We reject the joint test of $H_0: \beta_1 = 0, \beta_6 = 0$ at 5% level, as the p-value for the F test 0.014.

Interaction Terms

交叉项

- Effects of attendance on final exam score

$$= \Delta \text{stndfnl} / \Delta \text{atndrte} = \beta_1 + \beta_6 \text{priGPA}.$$

- ② The effect of atndrte on stndfnl is $-0.0067 + 0.0056(2.59) = 0.0078$, where $\text{mean}(\text{priGPA})$ is 2.59.
- A 10 percentage point increase in atndrte increases stndfnl by 0.0078 standard deviations from the mean final exam score.